

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS /R

Practical No: 13

Aim: Identifying and handling duplicates using distinct() (R).

Code:

```
library(dplyr)
```

```
# --- 1. SETUP: Load Data ---
```

```
df <- read.csv("Cancer.dataset.csv") %>%  
  select(id, diagnosis, radius_mean)
```

```
# 2. CHECK FOR DUPLICATES
```

```
duplicate_check <- df %>%  
  group_by(id) %>%  
  count() %>%  
  filter(n > 1)
```

```
print("--- 2. Duplicate Check Report ---")
```

```
if(nrow(duplicate_check) == 0) {  
  print("No duplicate Patient IDs found in the original file.")  
} else {  
  print(duplicate_check)  
}
```

```
# 3. HANDLING DUPLICATES (Example)
```

```
unique_diagnoses <- df %>%  
  distinct(diagnosis)
```

```
print("--- 3. Unique Diagnoses ---")
```

```
print(unique_diagnoses)
```

```
unique_radius <- df %>%
```

```
  distinct(radius_mean, .keep_all = TRUE)
```

```
print("--- 4. Unique Radius Measurements ---")
```

```
print(paste("Original rows:", nrow(df), "| Unique Radius rows:", nrow(unique_radius)))
```

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Output:

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for loading data, creating two datasets, and combining them vertically.
- Console:** Displays the output of the R code, including data structure checks, row counts, and a preview of the combined data.
- Environment:** Lists the objects created in the R session, such as `df`, `df_batch_1`, `df_batch_2`, and `combined_data`.

```
# R Script: Vertical Concatenation using rbind() / bind_rows()
> library(dplyr)
> # --- 1. SETUP: Load Data ---
> df <- read.csv("Cancer.dataset.csv") %>%
+   select(id, diagnosis, radius_mean, texture_mean)
> # --- 2. Create Two Separate Datasets ---
> df_batch_1 <- head(df, 5)
> df_batch_2 <- data.frame(
+   id = c(999001, 999002),
+   diagnosis = c("M", "B"),
+   radius_mean = c(15.50, 12.30),
+   texture_mean = c(20.10, 18.50)
+ )
> print("---- Data Structure Check ----")
[1] "---- Data Structure Check ----"
> print(names(df_batch_1))
[1] "id"      "diagnosis" "radius_mean" "texture_mean"
> print(names(df_batch_2))
[1] "id"      "diagnosis" "radius_mean" "texture_mean"
> # 3. VERTICAL COMBINATION
> combined_data <- bind_rows(df_batch_1, df_batch_2)
> print("---- Combined Data Summary ----")
[1] "---- Combined Data Summary ----"
> print(paste("Batch 1 rows:", nrow(df_batch_1)))
[1] "Batch 1 rows: 5"
> print(paste("Batch 2 rows:", nrow(df_batch_2)))
[1] "Batch 2 rows: 2"
> print(paste("Total Combined rows:", nrow(combined_data)))
[1] "Total Combined rows: 7"
> print("---- Preview (Bottom rows show the new batch) ----")
[1] "---- Preview (Bottom rows show the new batch) ----"
> print(tail(combined_data))
  id diagnosis radius_mean texture_mean
2  842517      M      20.57      17.77
3  84300903     M      19.69      21.25
4  84348301     M      11.42      20.38
5  84358402     M      20.29      14.34
6   999001     M      15.50      20.10
7   999002     B      12.30      18.50
```